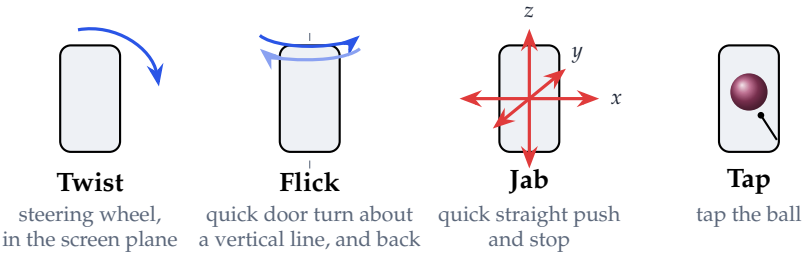
PhoneQubit: one qubit, one experiment

Superposition and collapse

Open `https://praveen2600.github.io/phonequbit/` in Safari or Chrome. Tap Start and allow motion. Turn on rotation lock, so the screen does not turn with the phone.



Twist
steering wheel, in the screen plane

Flick
quick door turn about a vertical line, and back

Jab
quick straight push and stop

Tap
tap the ball

This exercise needs only twist and tap. Flick and jab are for the other exercises.

The phone is a qubit. The ball shows the state: red is $|0\rangle$, blue is $|1\rangle$, purple is a mix. The angle θ is shown under the ball. A result appears on the ball and as "Measurement N gave ...".

- **Twist**: rotates the state. A quarter turn clockwise takes $|0\rangle \to \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.
- **Tap** the ball: measures. The result is 0 or 1.
- **Reset**: prepares $|0\rangle$. **Repeat**: prepares again the state from just before the last measurement.
- **Pause**: turning the phone no longer changes the state; taps still measure. **Go on** ends it.

Do

1. Hold the phone upright and tap **Go on** (gestures start paused). The ball is red, $\theta = 0°$. Tap it three times: always 0. Then tap **Reset** to start a new count.
2. Turn the phone a quarter turn clockwise, like a steering wheel, until it lies sideways. A short beep sounds, the ball turns purple, and $\theta = 90°$.
3. Tap **Pause**. Now turning the phone does nothing: hold it upright again, or lay it on the table. The state stays at $\theta = 90°$. Stay paused until the end; taps and **Repeat** still work.
4. **Predict:** out of 20 measurements of this state, how many give 1?
5. Tap the ball once and write the result in box 1. For boxes 2 to 20: tap **Repeat** (the ball turns purple again), then tap the ball once. At the end the screen shows "Measurement 20", and the Results line counts the 0s and 1s.
6. Now tap the ball once more, without Repeat. Compare with result 20.

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

Ones: _____

Expect

- Step 5: about 10 ones out of 20. Anything from about 6 to 14 is normal.
- Step 6: the same result as result 20.

Physics

After the twist the state is $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, a superposition. A tap measures it, with $P(1) = |\langle 1|\psi\rangle|^2 = \frac{1}{2}$. Each single result is random; only the frequency over many runs follows the rule (spread for 20 runs: $\sqrt{20 \cdot \frac{1}{2} \cdot \frac{1}{2}} \approx 2.2$).

The measurement also changes the state: after a result of 1 the qubit *is* $|1\rangle$ (collapse). That is why the extra tap repeats result 20, and why each new run needs **Repeat** to prepare a fresh copy of the superposition.

Going further. Reset, tap Go on, and switch to smooth angle (tap the "90° steps" chip under θ). Twist to any angle, tap **Pause**, and read θ. Predict $P(1) = \sin^2(\theta/2)$ and test it with 20 rounds of Repeat and tap. For $\theta = 60°$: about 5 ones in 20.

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